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Immersion cooling systems directing coolant flow between battery cells and cooling components internal and external to battery packs

Abstract

A battery pack is disclosed and includes battery cells, an outer housing, a support structure, and a planar structure. The outer housing is configured to enclose and immerse the plurality of battery cells in a coolant. The outer housing includes multiple walls including side walls, a top wall and a bottom wall. The support structure supports the battery cells. The support structure includes dividing members separating the battery cells. The planar structure is disposed above or below the battery cells. A gap exists between a planar surface of the planar structure and a planar surface of one of the top wall and the bottom wall of the outer housing for flowing the coolant between the planar structure and the one of the top wall and the bottom wall. The planar structure and at least some of the walls direct the coolant to flow across and between the battery cells.

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Background/Summary

INTRODUCTION

(1) The information provided in this section is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

(2) The present disclosure relates to battery packs of electric vehicles.

(3) An electric vehicle includes a battery pack that provides power to various vehicle systems including propulsion systems, lighting systems, infotainment systems, air-conditioning systems, braking systems, steering systems, autonomous control systems, navigation systems, etc. The propulsion system may include one or more motors for driving wheels of the vehicle.

(4) A battery pack typically includes an outer housing that encloses the battery pack. Power is supplied from the battery pack through the housing to various vehicle systems and devices. A control circuit may be disposed on or external to the housing and monitor states of the battery pack and control supply of power to the vehicle systems.

SUMMARY

(5) A battery pack is disclosed and includes battery cells, an outer housing, a support structure, and a planar structure. The outer housing is configured to enclose and immerse the battery cells in a coolant. The outer housing includes multiple walls including side walls, a top wall and a bottom wall. The support structure supports the battery cells. The support structure includes dividing members separating the battery cells. The planar structure is disposed above or below the battery cells. A gap exists between a planar surface of the planar structure and a planar surface of one of the top wall and the bottom wall of the outer housing for flowing the coolant between the planar structure and the one of the top wall and the bottom wall. The planar structure and at least some of the walls direct the coolant to flow across and between the battery cells.

(6) In other features, the planar structure and the top wall direct the coolant across tops of the battery cells prior to or subsequent to flowing between the battery cells.

(7) In other features, the planar structure and the bottom wall direct the coolant across bottoms of the battery cells prior to or subsequent to flowing between the battery cells.

(8) In other features, the planar structure is configured as a splitter and receives coolant and splits the received coolant into flow paths to direct the coolant around sides of each of the battery cells.

(9) In other features, the planar structure directs the coolant to flow vertically between and around each of the battery cells.

(10) In other features, the planar structure is configured as a manifold and receives coolant from each of the battery cells and direct the coolant to the gap between the planar structure and the one of the top wall or the bottom wall.

(11) In other features, the battery pack further includes an electrical bus disposed across and electrically coupled to the battery cells. The support structure is configured to hold the electrical bus. The support structure is configured to direct the coolant across a surface of the electrical bus.

(12) In other features, the planar structure includes holes for passage of the coolant to flow to or from each of the battery cells. The support structure includes feet, where open areas between the feet are respectively over the holes for passage of the coolant to flow to or from each of the battery cells.

(13) In other features, gaps between the feet allow the coolant to flow between the feet and along sides of the battery cells.

(14) In other features, the planar structure includes first holes for passage of the coolant to flow to or from each of the battery cells. The support structure includes second holes. The first holes are larger than the second holes. Portions of the support structure including the second holes are disposed within the first holes.

(15) In other features, the support structure includes orifices are arranged for flow of the coolant to or from the battery cells. Each portion of the support structure surrounding a respective one of the orifices is formed of a material that melts during a thermal runaway event of a corresponding one of the battery cells.

(16) In other features, the battery pack further includes the support structure including protruding elements extending in gaps between the battery cells.

(17) In other features, the protruding elements include at least one of spiral-shaped elements, disks, rods, or trees.

(18) In other features, a cooling system is disclosed and includes a battery pack, a pump, and a control module. The pump is configured to pump coolant to and from the battery pack. The control module is configured to control the pump to circulate the coolant through the battery pack.

(19) In other features, a cooling system is disclosed and includes: the battery pack; at least one of a

cell monitoring module, a control module, a battery disconnect module, or a motor; and conduits connected to and directing the coolant between the battery pack and the at least one of the cell monitoring module, the control module, the battery disconnect module, or the motor.

(20) In other features, a cooling system is disclosed and includes the battery pack and at least one of a cell monitoring module, a control module or a battery disconnect module disposed on a wall of the battery pack and contacting the coolant in the battery pack.

(21) In other features, a thermal control system is disclosed and includes: a cooling circuit including the battery pack and a swirl pot, where the swirl pot is configured to perform fluid deaeration; a sensor coupled to the swirl pot, the battery pack or another component of the cooling circuit and configured to monitoring a state of the thermal control system; and a control module is configured to detect a thermal runaway event based on an output of the sensor and, perform a countermeasure in response to detecting the thermal runaway event.

(22) In other features, a battery pack is disclosed and includes battery cells and an outer housing. The battery cells include first cells and second cells, where: the first cells are arranged in a first set of rows; the second cells are arranged in a second set of rows; and each of the second set of rows are disposed between and laterally offset from two of the first set of rows. The outer housing is configured to enclose and immerse the battery cells in a coolant, the outer housing including a first wall and a second wall, where: the first wall includes first orifices each of which directing a portion of the coolant to flow in a row-to-row manner from the first wall to the second wall; the second wall is disposed on an opposite side of the battery cells as the first wall and includes second orifices to receive the coolant subsequent to flowing between the battery cells; a first row of the first set of rows is disposed closer to the first wall than the other rows of the first set of rows and closer to the first wall than the second set of rows; and a last row of the second set of rows is disposed closer to the second wall than the other rows of the second set of rows.

(23) In other features, the battery pack further includes a support structure for supporting the battery cells and including dividing members separating the battery cells, where: the battery cells are arranged such that adjacent ones of the first cells are separated by a portion of the support structure and a portion of the coolant and not by one or more of the second cells; the second wall is arranged parallel to the first wall; the first wall is a front wall having an outer planar surface facing in a vehicle forward direction when the battery pack is installed in a vehicle; and the second wall is a back wall having an outer planar surface facing in a vehicle rearward direction when the battery pack is installed in the vehicle.

(24) In other features, a battery pack is disclosed and includes battery cells including first cells and second cells, where: the first cells are arranged in a first set of rows; the second cells are arranged in a second set of rows; and each of the second set of rows are disposed between and laterally offset from two of the first set of rows. The battery pack further includes an outer housing configured to enclose and immerse the battery cells in a coolant. The outer housing includes: a first wall including first orifices each of which directing a portion of the coolant to flow in across rows in a column-by-column manner; and a second wall disposed on an opposite side of the battery cells as the first wall and including second orifices to receive the coolant subsequent to flowing between the battery cells. A first cell of each of the first set of rows and a first cell of each of the second set of rows are disposed closer to the first wall than other cells of the first set of rows and other cells of the second set of rows. A last cell of each of the first set of rows and a last cell of each of the second set of rows are disposed closer to the second wall than other cells of the first set of rows and other cells of the second set of rows.

(25) Further areas of applicability of the present disclosure will become apparent from the detailed description, the claims and the drawings. The detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:
- (2) FIG. 1 is a functional block diagram of an example of a vehicle power system including an immersion cooling system in accordance with the present disclosure;
- (3) FIG. 2 is a functional block diagram of a vehicle including a power system including a power source with a battery pack in accordance with the present disclosure;
- (4) FIG. 3 is a functional block diagram of an example of an immersion cooling system including a side cross-sectional view of a battery pack in accordance with the present disclosure;
- (5) FIG. 4 is a side cross-sectional view of an example of a battery pack including a bottom mounted planar cooling and distribution structure in accordance with the present disclosure;
- (6) FIG. 5 is a side cross-sectional view of an example of a battery pack including a top mounted planar cooling and distribution structure in accordance with the present disclosure;
- (7) FIG. 6 is a side cross-sectional view of an example of a battery pack including a top mounted planar cooling and distribution structure and an intake with side holes in accordance with the present disclosure;
- (8) FIG. 7 is a cross-sectional side view of an example of a battery pack including a bottom mounted planar structure implemented as a splitter and illustrating bottom up flow of immersion fluid in accordance with the present disclosure;
- (9) FIG. 8A is a cross-sectional side view of an example of a portion of a support structure, splitter and battery cell of the battery pack of FIG. 7 illustrating orifices of the support structure and the planar structure in accordance with the present disclosure;
- (10) FIG. 8B is a cross-sectional bottom view at section line A-A of FIG. 8A;
- (11) FIG. 8C is a cross-sectional view at section line B-B of FIG. 8A;
- (12) FIG. 8D is a cross-sectional top view at section line C-C of FIG. 8A;
- (13) FIG. 9A is a cross-sectional side view of an example of a portion of a support structure, splitter and battery cell of a battery pack illustrating a protruding portion of the support structure in accordance with the present disclosure;
- (14) FIG. 9B is a cross-sectional bottom view at section line D-D of FIG. 9A;
- (15) FIG. 10A is a cross-sectional view of the portion of the support structure, the splitter and the battery cell of FIG. 9A illustrating a melted portion of the support structure due to a thermal runaway event;
- (16) FIG. 10B is a cross-sectional view at section line E-E of FIG. 10A illustrating a melted away area of the support structure;
- (17) FIGS. 11A-B are perspective views of an example of a portion of a support structure and corresponding battery cells illustrating some protruding elements of the support structure in accordance with the present disclosure;
- (18) FIG. 12 is a top cross-sectional view of an example of a battery pack including a forward disposed splitter and a rearward disposed manifold for directing coolant between battery cells of the battery pack in a direction parallel to rows of the battery cells in accordance with an embodiment of the present disclosure;
- (19) FIG. 13 is a top cross-sectional view of an example of a battery pack including a forward disposed splitter and a rearward disposed manifold for directing coolant between battery cells of the battery pack in a direction perpendicular to rows of the battery cells in accordance with an embodiment of the present disclosure; and
- (20) FIG. 14 illustrates a thermal runaway event detection method in accordance with the present disclosure.

(21) In the drawings, reference numbers may be reused to identify similar and/or identical elements.

DETAILED DESCRIPTION

(22) Cooling high power-density battery packs during energy-intrusive conditions such as during direct current (DC) fast-charging events and high-vehicle performance events can be difficult. The high-performance events refer to events during which a large amount of power is quickly and iteratively being discharged and regenerated. Many elements within a battery pack are designed to have (i) low electrical conductivity for electrical isolation which typically also restricts heat flow, and/or (ii) to provide protection by restricting heat transfer during a thermal runaway event. A battery cell can have low conductivity, especially through a stack of cathodes, separators, and anodes, which makes heat removal from the largest surfaces of the battery cell difficult. Battery cells typically have a low maximum temperature to prevent accelerated degradation, which limits the temperature difference required for heat transfer.

(23) Immersion cooling, where the battery cells are directly submerged in a circulated dielectric fluid, can enable high amounts of heat to be carried away from battery cells of a battery pack. Immersion cooling immerses the battery cells in an immersion fluid (or coolant), which maximizes total contact surface area for heat transfer and eliminates many limitations imposed on traditional cold plates and ribbons. Traditional cold plates (planar cooling structures) and ribbons are formed of a metallic material and have a dielectric coating (or outer layer). The cold plates may be disposed below battery cells and are used to absorb thermal energy from the battery cells. The ribbons are wavy bands that extend between and adsorb thermal energy from the battery cells, which are cylindrically shaped. The dielectric coating (or outer layers) of the cold plates and ribbons limit the amount of cooling provided. The cold plates and the ribbons have limited ability to cool battery cells during energy-intensive conditions, which can result in the battery cells increasing in temperature, degrading, experiencing thermal runaway events, and/or having reduced life expectancy.

(24) Examples set forth herein include immersion cooling systems include support structures, planar cooling and distribution structures, splitters, manifolds, and protruding elements that aid in directing flow of an immersion fluid (or coolant) to maximize contact cooling surface area. The planar structures may refer to structures having planar support and/or contact surfaces for supporting and/or contacting, for examples cells and/or buses. This includes control of passive and/or active coolant flow through components internal and external to the battery packs. Cooling circuits are disclosed that circulate coolant between battery packs and components external to the battery packs. In addition, examples are provided for direct contact cooling of modules mounted on battery packs by directing coolant to circulate by and contact exterior surfaces of the modules. The immersion cooling systems improve overall cooling and operating performance, cost and manufacturability of battery packs and components external to and in the same cooling circuits as the battery packs.

(25) FIG. 1 shows a vehicle power system **100** including an immersion cooling system **102**, a vehicle control module **104**, one or more motors **105**, and other loads **106**. The immersion cooling system **102** includes a battery pack **108**, a cell monitoring module **110**, a battery disconnect module **112**, the one or more motors **105**, a swirl pot **114**, a pump **116** and a heat exchanger **118**. The pump **116** circulates an immersion fluid (or coolant) through a cooling circuit **120**. The coolant is a non-conductive (or dielectric) coolant and provides isolation between adjacent cells of the battery pack **108**. As an example, the cooling circuit **120** may include the one or more motors **105**, the battery pack **108**, the cell monitoring module **110**, the battery disconnect module **112**, the swirl pot **114** and the heat exchanger **118** and conduits fluidly connecting these components. Although the battery disconnect module **112** is shown separate from the battery pack **108**, the battery disconnect module **112** may be attached to or incorporated in the battery pack **108**.

(26) Although certain components are shown as part of the cooling circuit **120**, other components

may be included. The coolant may be circulated through and cool one or more other components. Also, although the cooling circuit components are shown in a certain serially connected order, the arrangement of the components in the cooling circuit may be different from that shown in FIG. 1. For example, the circuit components may be connected in a different order and/or two or more of the circuit components may be connected in parallel. In an embodiment, the heat exchanger **118** is connected between the pump **116** and the battery pack **108** to provide the coolest temperature coolant to the battery pack **108**. In another embodiment, the pump **116** is connected between the swirl pot **114** and the one or more motors **105**. In another embodiment, the one or more motors **105** are connected between the battery pack **108** and the heat exchanger **118**. The heat exchanger **118** may be located near a front grill of the host vehicle.

(27) The vehicle control module **104** may be in communication with the modules **110**, **112** and control operation of the loads **106**. The one or more motors **105** may be used for propulsions purposes and drive one or more wheels of a vehicle. Examples of the loads **106** are shown in FIG. 2 and may include electronic components, devices, and systems including propulsion motors, actuator motors, lighting, navigation devices, infotainment devices, etc. The loads **106** may include high-voltage loads and low-voltage loads.

(28) The battery pack **108** may be configured similarly as any of the battery packs disclosed herein. Some of which are shown in FIGS. 3-11. The battery packs shown in FIGS. 3-11 may be referred to as battery modules. Any number of the battery modules may be included in a battery pack and/or in a battery tray.

(29) The cell monitoring module **110** monitors operation and states of, for example, battery cells of the battery pack **108** and controls operation of the battery pack **108**. The cell monitoring module **110** may monitor states-of-charge and temperatures of cells and cell group voltages. The cell monitoring module **110** is responsible for balancing parallel cell groups to make sure that each group has the same voltage by selectively discharging high groups.

(30) The battery disconnect module **112** may, based on the states of the battery cells of the battery pack **108**, state of the cooling circuit **120**, and/or other states and/or conditions disconnect one or more of the motors **105** and the other loads **106** from receiving power from the battery pack **108**. The battery disconnect module **112** may control distribution of power to the motors **105** and the other loads **106**. In one embodiment, two or more of the modules **104**, **110**, **112** are integrated into a single module.

(31) The swirl pot **114** may be disposed at the high elevation point of the cooling circuit **120** and deaerates the coolant. The swirl pot **114** may also regulate pressure in the cooling circuit **120** by for example venting gases when pressure in the cooling circuit **120** exceeds a threshold. The swirl pot **114** may also be a point where the cooling circuit **120** is filled with coolant. Coolant may flow in a circular manner through the swirl pot **114** to remove air (or gas) from the coolant. Sensors **122** may be connected to and/or included within the swirl pot **114**. In addition or as an alternative, sensors **124** may be included in the battery pack **108**. The sensors **122**, **124** may include pressure sensors, chemical species sensors, etc. for detecting thermal runaway events. The species sensors may be used to detect gases, such as carbon dioxide, carbon monoxide, hydrocarbon gas, etc. and/or other gases released during a thermal runaway event. The sensors **124** may also be used for detecting temperatures, voltages, current levels, etc. of the battery cells of the battery pack **108**. Any of the modules **104**, **110**, **112** may monitor the states of the sensors **122**, **124** and determine based on this information whether to isolate and/or disable one or more of the battery cells, perform cell balancing, adjust coolant flow rate and/or pressure, and/or perform other operations.

(32) The swirl pot **114** may be used as a central monitoring location to detect a thermal runaway event anywhere within the cooling circuit **120** via pressure monitoring, combustion species detections, and/or monitoring of another parameter. When a thermal runaway event occurs, certain gases are released from one or more cells and a pressure within the cooling circuit **120** increases. These gases and pressure may be detected anywhere in the cooling circuit **120**. The gases vented

from the one or more cells are released into the coolant. The swirl pot **114** is at the highest location in the cooling circuit **120** and may include a pressure vent for releasing the gases from the cooling circuit **120**. Pressure sensing may be used to detect the event since the coolant is incompressible and cell venting produces a fast pressure wave propagation event through the cooling circuit **120**, which is detected by, for example, a centrally located pressure sensor. In one embodiment, two pressures sensors are included for redundancy/validation purposes. As an example, one pressure sensor may be located in the swirl pot **114** and another pressure sensor may be located in the battery pack **108**. Although pressure, species and/or other sensors may be located at each cell in the battery pack **108**, to minimize complexity, mass and costs, a single pressure sensor and gas species sensor may be located in the battery pack **108**. In one embodiment, two species sensors are included, one in the swirl pot **114** and another in the battery pack **108**.

(33) The heat exchanger (or chiller) **118** draws thermal energy out of the coolant prior to the coolant flowing between and around the battery cells of the battery pack **108**.

(34) FIG. 2 shows a vehicle **200** including a power system **202** including a power source **204**. The power source **204** includes the battery pack **108** and the cell monitoring module **110**. The vehicle **200** may also include the vehicle control module **104**, an infotainment module **206** and other control modules **208**. The cell monitoring module **110** and the modules **204**, **206**, **208** may communicate with each other via one or more buses **210**, such as a controller area network (CAN) bus and/or other suitable interfaces. The vehicle control module **104** may control operation of vehicles systems. The vehicle control module **104** may include a mode selection module **212**, a parameter adjustment module **214**, as well as other modules. The mode selection module **212** may select a vehicle operating mode. The parameter adjustment module **214** may be used to adjust parameters of the vehicle **200**.

(35) The vehicle **200** may further include: a memory **218**; a display **220**; an audio system **222**; one or more transceivers **223** including sensors **226**; and a navigation system **227** including a global positioning system (GPS) receiver **228**. The sensors **226** may include sensors, cameras, objection detection sensors, temperature sensors, accelerometers, vehicle velocity sensor, and/or other sensors. The GPS receiver **228** may provide vehicle velocity and/or direction (or heading) of the vehicle and/or global clock timing information.

(36) The memory **218** may store sensor data **230**, vehicle parameters **232**, cooling circuit information **234** (e.g., battery pack or cooling circuit parameters and/or states), and applications **236**. The applications **236** may include applications executed by the modules **204**, **206**, **208**. Although the memory **218** and the vehicle control module **104** are shown as separate devices, the memory **218** and the vehicle control module **104** may be implemented as a single device.

(37) The vehicle control module **104** may control operation of: a propulsion system **240** including electric motors **242**; a window/door system **250**; a lighting system **252**; a seating system **254**; a mirror system **256**; a brake system **258**; and/or a steering system **262** according to parameters set by the modules **104**, **110**, **206**, **208** and/or the cooling circuit information **234**. The vehicle control module **104** may set some of the parameters based on signals received from the sensors **226**. The vehicle control module **104** may receive power from the power source **204**, which may be provided to the motors **242**, the window/door system **250**, the lighting system **252**, the seating system **254**, the mirror system **256**, the brake system **258** and/or the steering system **262**, etc. Some of the vehicle control operations may include starting and/or controlling running operation of the electric motors **242**, unlocking doors of the window/door system **250**, powering any of the systems **250**, **252**, **254**, **256**, **258**, **262**, and/or performing other operations.

(38) The window/door system **250**, the lighting system **252**, the seating system **254**, the mirror system **256**, the brake system **258** and/or the steering system **262** may include actuators controlled by the vehicle control module **104** to, for example, adjust steering wheel angle, throttle position, pedal position, door locks, window position, seat angles, etc. This control may be based on the outputs of the sensors **226**, the navigation system **227**, the GPS receiver **228** and the above-stated

data and information stored in the memory **218**.

(39) The vehicle control module **104** may determine various parameters including a vehicle speed, a motor speed, a motor torque, a gear state, an accelerometer position, a brake pedal position, an amount of regenerative (charge) power, an amount of boost (discharge) power, an amount of auto start/stop discharge power, and/or other information, such as priority levels of source terminals of the power source **204**, power, current and voltage demands for each source terminal, etc. The vehicle control module **104** may share this information and the vehicle operating mode with the cell monitoring module **110** and/or the battery disconnect module **112** of FIG. **1**. The cell monitoring module **110** may determine other parameters, such as: an amount of charge power at each source terminal of the battery pack **108**; an amount of discharge power at each source terminal of the battery pack **108**; maximum and minimum voltages at source terminals of the battery pack **108**; maximum and minimum voltages at power rails, battery cells, modules, etc. of the battery pack **108**; SOX values of battery cells and/or modules; temperatures of battery cells and/or modules; current values of battery cells and/or modules; power values battery cells and/or modules; etc. SOX is a generic term for state of charge (SOC), state of health (SOH), and state of power (SOP). The cell monitoring module **110** may determine connected configurations of the battery cells and corresponding switch states as described herein based on the parameters determined by the vehicle control module **104**, the cell monitoring module **110** and/or the battery disconnect module **112** of FIG. **1**. The above-stated information and parameters may be shared by the modules **104**, **110**, **112**.

(40) FIG. **3** shows an immersion cooling system **300** including a battery pack **302**, a cell monitoring module **304**, a battery disconnect module **306** and a motor **308**. The immersion cooling system **300** may be part of a cooling circuit and/or a larger immersion cooling system, such as that shown in FIG. **1**.

(41) The battery pack (or module) **302** includes an outer housing **310**, a support structure (referred to as a support frame) **312**, battery cells (referred to simply as cells) **314**, a electrical bussing **316**, an inlet **318** and an outlet **320**. The outer housing **310** includes a top wall **330**, a bottom wall **332**, and side walls **334**. The support structure **312** includes a lower matrix support member **340** and an upper matrix support member **342** and may include one or more connecting members **344** extending between the members **340**, **342**. The matrix support members **340**, **342** include dividers **346**, **348** that separate the battery cells **314**. The battery cells **314** may be “can” cells that are cylindrically-shaped. The support structure **312** holds the cells **314** in place. The cells **314** sit on the lower matrix support member **340**. The bus **316** is supported by the upper matrix support member **342**, which holds top portions of the cells **314** in place.

(42) Non-conductive immersion fluid (or coolant) fills the outer housing **310** and is passed from the inlet **318** down and then laterally across the cells **314** in a serial manner. The coolant is then directed upward and over the top of the bus **316** to the outlet **320**. Arrows show an example direction of coolant flow; other flow patterns may be implemented. The coolant is in contact with the cell monitoring module **304** and the bus **316** and thus cools the cells **314**, the cell monitoring module **304** and the bus **316**. The bus **316** and other buses referred to herein may be formed of a conductive material, such as copper and/or aluminum and provides connections between the cells **314** and the cell monitoring module **304**. Although one bus is referred to with regards to the embodiment of FIG. **3** and some other embodiments, each embodiment may include any number of buses.

(43) FIG. **4** shows a battery pack (or battery module) **400** that includes an outer housing **410**, a support structure (referred to as a support frame) **412**, battery cells (referred to simply as cells) **414**, a bus **416**, an inlet **418** and an outlet **420**. A planar cooling structure (or portion) **422** of the support structure **412** performs in combination with a bottom wall **424** of the outer housing **410** to provide a planar fluid flow passage of non-conductive immersion fluid (or coolant) along the bottom of the support structure **412** in a first direction, up and then laterally between the cells **414** in an opposite

direction, as shown. Arrows show an example direction of coolant flow; other flow patterns may be implemented. In this manner, bottoms of the cells **414** is cooled as well as side surfaces of the cells **414**. The coolant fills the outer housing **410** and may flow in an opposite direction than shown. In the example shown, the bus **416** is disposed on the top of the outer housing **410** and may be supported by tops of the cells **414**. In another embodiment, the cells **414** and the bus **416** are entirely enclosed in the outer housing **410** and contact and are cooled by the coolant.

(44) The arrangement between the planar cooling structure **422** and the bottom wall **424** define cooling flow to ensure module-level flow homogeneity as well as bottom cooling of the cells **414**. The coolant is directed to the extremities of the battery pack **400** while cooling the bottoms of the cells **414**. Coolant flow then returns around the outside faces of the cells **414** for greater surface contact area to increase heat transfer before returning to the outlet **420**, which is located adjacent the inlet **418**. The planar cooling structure **422** and/or the like may be located on top of the cells **414** and/or the bus **416** for cooling tops of the cells **414** and/or the bus **416**. The inlet **418** may have a single bottom output, as shown, or multiple side hole outputs depending on flow direction of the coolant. Other examples are shown in FIGS. 5-7.

(45) FIG. 5 shows a battery pack (or battery module) **500** that includes an outer housing **510**, a support structure (referred to as a support frame) **512**, battery cells (referred to simply as cells) **514**, a bus **516**, an inlet **518** and an outlet **520**. A planar cooling structure (or portion) **522** of the support structure **512** performs in combination with a top wall **524** of the outer housing **510** to provide a planar fluid flow passage of non-conductive immersion fluid (or coolant) along the top of the support structure **512** in a first direction, down and then laterally between the cells **514** in an opposite direction, as shown. Arrows show an example direction of coolant flow; other flow patterns may be implemented. The portion **522** may include a plate **526**, which may be in direct contact with the bus **516** and cool the bus as well as tops of the cells **514**. In one embodiment, the plate **526** is not included and the coolant directly contacts the top of the bus **516**. The coolant fills the outer housing **510** and may flow in an opposite direction than shown. In the example shown, the bus **516** is disposed on tops of the cells **514** and may be supported by the support structure **512**.

(46) FIG. 6 shows a battery pack (or battery module) **600** that an outer housing **610**, a support structure (referred to as a support frame) **612**, battery cells (referred to simply as cells) **614**, a bus **616**, an inlet **618** and an outlet **620**. The battery pack **600** includes the inlet **618** with side holes **621** for directing non-conductive immersion fluid (or coolant) to side surfaces of the cells **614**. A plate (or portion) **622** of the support structure **612** performs in combination with a top wall **624** of the outer housing **610** to provide a planar fluid flow passage of non-conductive immersion fluid (or coolant) along the top of the support structure **612**. In the example shown, the fluid flows from the holes **621** laterally between the cells **614**, up and then in an opposite direction across the top of the support structure to the outlet **620**, as shown. Arrows show an example direction of coolant flow; other flow patterns may be implemented. The portion **622** may include a plate **626**, which may be in direct contact with the bus **616** and cool the bus as well as tops of the cells **614**. In one embodiment, the plate **626** is not included and the coolant directly contacts the top of the bus **616**. The coolant fills the outer housing **610** and may flow in an opposite direction than shown. In the example shown, the bus **616** is disposed on tops of the cells **614** and may be supported by the support structure **612**. The inlet **618** extends down a side of one of the cells to a sidewall **630**, which then extends down to a bottom wall **632**.

(47) FIG. 7 shows a battery pack (or battery module) **700** including a bottom side planar cooling structure **702** implemented as a flow distribution manifold. The planar cooling structure **702** receives non-conductive immersion fluid (or coolant) and includes holes (some of which designated **704**) that are centered with bottom surfaces of cells (some of which designated **706**). In this example shown, the immersion fluid flows upward through the holes **704** and directed at the cells **706**. The immersion fluid flows in an upward direction around sides of each of the cells **706** and contacts sides surfaces of the cells **706**. Coolant flows into the planar cooling structure **702**,

through the holes **704**, upward around the cells **706** and then to an upper portion **708** of the battery pack **700** and out the upper portion **708**. The upper portion **708** performs as a manifold.

(48) FIG. 7 shows a “close up” portion **730** of the battery pack **700** to further illustrate a corresponding support structure **732** and coolant flow. The support structure **732** includes an upper portion (or matrix) **734** and a lower portion (or matrix) **736**. The upper portion **734** includes dividers **738** that extend between the cells **706** and hold top portions of the cells **706** in place. The lower portion **736** includes pillars **740** that extend between the cells **706** from middle hexagonally-shaped dividers **737**, which are between middle portions of the cells, down to lower hexagonally-shaped dividers **739**. The lower-hexagonally-shaped dividers **739** are above lower structure feet **743**. The dividers **737**, **739**, the pillars **740**, and the lower structure feet **743** may be integrally formed as a single component.

(49) The pillars **740** may include protruding members **745** that extend into holes **741** in a bottom planar structure **702**. Gaps between the pillars **740** and the cells **706** allow coolant to flow around the cells. The pillars **740** interlock with the planar structure **702**. In the shown embodiment, the lower matrix **736** includes the separated lower structure feet **743** that are integrally formed with ends of the pillars **740**, as shown. The pillars **740** extend into a top plate **744** of the planar structure **702**. The pillars **740** and holes **741** align and center holes **704** of the planar structure **702** to gaps and open area between the lower structure feet **743** (some of the gaps are represented by arrows **746** and the open area are designated as **747**, as seen in FIG. 8B) and to centers of bottom surfaces of the cells **706**. Coolant passes through the holes **704**, around the cells **706**, in the gaps **746** and open area **747**, and up through holes **750** in the dividers **738** located at least partially between the cells **706**.

(50) The holes **704** provide direct fluid impingement on bottom faces of the cells **706** for improved heat transfer coefficient (HTC) and flow balancing. Flow along the sides of the cells **706** provides increased heat transfer from the cells **706** to the coolant. Bottom-to-top flow aids in deaeration of the fluid flow and decreases pumping work because fluid naturally tends to rise with increase in temperature. In an embodiment, the support structure **732** is formed from a plastic material and configured as a matrix for cell positioning, structural supporting and electrical isolating purposes. The support structure **732** may be injection molded in one embodiment. Example views of portions of the matrix are shown in FIGS. 8B-D. Although bottom-to-top coolant flow is shown, coolant may flow in an opposite direction. The support structure **732** also spaces the cells **706** off the planar structure **702** to provide a fluid path and electrical isolation. The support structure **732** is also configured to direct the coolant to flow along the sides of the cells **706** and decrease the flow area to increase fluid flow velocity near the cells **706**.

(51) The holes **704** create jets of fluid that impinge directly on the bottom surfaces of the cells **706**, increasing HTCs of the bottom surfaces. The sizes of the holes **704** may vary across the battery pack **700**. For example, holes closer to the center of the battery pack or planar structure **702** may be smaller than holes further from the center of the battery pack **700** or planar structure **702**. The center of the battery pack **700** is shown by centerline **760**, which also passes through the center of the planar structure **702**. This variance in hole sizes (i.e., diameters and/or cross-sectional area of the holes) aids in providing a more balanced and/or uniform flow of coolant around the cells **706**. Coolant flowing out of the center of the upper portion (or manifold) **708** and may be provided to one or more heat exchangers, one or more chillers, one or more pumps, and/or to one or more other components to be cooled.

(52) In the example shown, a bus **770** is disposed on the support structure **732**. The upper portion **734** of the support structure **732**, which may be connected to or separate from the lower portion **736**, provides an insulated member on which to mount the bus **770**.

(53) FIG. 8A shows a portion **800** of the support structure **732** and the planar cooling structure **702** and one of the battery cells **706** of the battery pack **700** of FIG. 7. FIG. 8B shows a cross-sectional bottom view at section line A-A of FIG. 8A. The lower matrix **736** include the feet **743**, which may

be triangular-shaped, as shown, and pillars **740** that extend into the top plate **744** of the planar structure **702**. The hole **704** is centered between the feet **743** below the cell **706**. In FIG. **8B**, the hole **704** is shown by a dashed circle to illustrate where the hole **704** is located, but in actuality is not in the cross-sectional view of FIG. **8B**. FIG. **8C** shows a cross-sectional view at section line B-B of FIG. **8A**. The dividers **737** are shown. FIG. **8D** shows a cross-sectional view at section line C-C of FIG. **8A**. The upper matrix **734** is shown in FIG. **8D**.

(54) The pillars **740** include the protruding elements **745** (also referred to as locating features). The locating features ensure alignment and structural integrity when in use. In one embodiment, the lower portion **736** is aligned and disposed on the planar structure **702**. The cells **706** are then inserted into the lower portion **736** between the dividers **740**. The upper portion **734** of the support structure **732** is then fitted on the cells **706** and may or may not be connected to the lower portion **736**. The bus **770** may then be added and connected via conductive elements (e.g., wires) to the cells **706**.

(55) FIG. **9A** shows a portion **900** of a support structure **902**, similar to the support structure, **732** of FIG. **7**. However, the support structure **902** includes a lower matrix **903**, which may be integrally formed as part of the support structure **902**, with (i) protruding members (one protruding member **904** is shown) that extend into large holes (one hole **906** is shown) of a planar structure **908**, and (ii) triangular-shaped feet **905**. The protruding members **904** of the lower matrix **903** are placed into the large holes **906**. The protruding members **904** includes flow-control holes (one hole **910** is shown) through which non-conductive immersion fluid (or coolant) is passed. The support structure **902** centers battery cells (one battery cell **912** is shown) over the holes **910**. FIG. **9B** shows a cross-sectional bottom view at section line D-D of FIG. **9A**.

(56) FIG. **10A** shows the portion **900** of the support structure **902** illustrating a melted portion **1000** of the support structure **902** due to a thermal runaway event. FIG. **10B** shows a cross-sectional view at section line E-E of FIG. **10A** illustrating a melted away area **1002** of the support structure **902**. The support structure **902** or lower portions thereof (e.g., the protruding member **1002**) is formed of a material (e.g., plastic) having material properties that have a low melting point. The protruding member **1002** provides a flow control orifice that is located in a path of thermal runaway below a cell **912**. A gas vent is built into the cell **912** and will open if the cell **912** experiences a thermal runaway event. When it opens, it melts a portion of member **1002**, which allows more cooling flow to go to the cell **912** during the thermal runaway event. Hot gases from the cell **912** are designated as **1004**. The hot gases **1004** melt away the material and enlarge the hole **910** providing greater immersion fluid flow to the cell **912** to increase cooling of the cell and prevent thermal runaway events from occurring in cells adjacent to the cell **912**. This provides a passive implementation for biasing coolant flow to the cell **912** during a thermal runaway event. The hot gases are ejected out of the bottom of the cell **912**, directed towards the hole **910** and passed to the coolant, which are then spread into the coolant and cooled by the coolant. The coolant passes through the cell **912** and neighboring cells and returned to, for example, the swirl pot **114** of FIG. **1**, where the gases are then able to be released from the corresponding coolant circuit.

(57) FIGS. **11A-B** shows a portion **1100** of a support structure and corresponding battery cells **1102** illustrating some protruding elements **1104** of the support structure that could be implemented as part of the pillars **740** of FIG. **7** or elsewhere. Each of the protruding elements **1104** extend from a lower matrix **1106** of the support structure and in open spaces (or gas) between three adjacent cells **1102**. The gaps, in the example shown, are triangular-shaped, where sides of each triangle are curved and match outer dimensions of the cells **1102**. Although the protruding elements **1104** are shown as spirals, the protruding elements may be implemented as rods with attached rings, trees with branches, etc. The protruding elements **1104** provide fluid flow resistance and are used to change local fluid flow direction to increase heat transfer coefficients for outside surfaces of the cells **1102** while providing additional structural support and electrical isolation. Tops of the protruding elements **1104** may be attached to an upper matrix. The protruding elements **1104** may

be molded with the lower matrix **1106** to form a single part. The lower matrix **1106** may include rings **1108** as shown on which the cells **1102** are disposed. The rings **1108** may be spaced away from a bottom plate **1110**. Coolant may pass between the plate **1110** and the rings **1108**, up through the rings **1108** and around the cells **1102**. Arrows in FIG. **11B** show direction of coolant flow. The rings **1108** may be connected to the plate **1110** via support members not shown in FIGS. **11A-11B** and/or may be connected to side walls not shown in FIGS. **11A-11B**.

(58) FIG. **12** shows a battery pack **1200** including a forward disposed splitter **1202** and a rearward disposed manifold **1204** for directing coolant between battery cells **1206** of the battery pack **1200** in a direction parallel to rows of the battery cells **1206**. The splitter **1202** has holes **1208** that vary in size from a centrally located inlet **1210**. The holes **1208** may increase in size the further the holes **1208** are away from the inlet **1210**, as shown. The manifold **1204** may have similar holes **1211** and an outlet **1212**.

(59) The cells **1206** are arranged in rows and columns, where the rows extend parallel to the splitter **1202** and the manifold **1204** and perpendicular to the direction of immersion fluid flow, which is from the splitter **1202** to the manifold **1204**. The rows include first rows that are offset from the second rows to provide a close arrangement of cells. A winding arrow is shown to illustrate direction and flow of the immersion fluid.

(60) The example of FIG. **12** provides fore-to-aft cooling to minimize the number of cells cooled in series. Foremost and aftmost rows of cells are arranged along long surfaces of the splitter and manifold of the battery pack (or battery module). Orifices of the splitter and manifold are sized to control flow distribution. Orientation of the rows of cells affect heat transfer and pressure drop between the inlet and the outlet. Centrally locating the inlet and outlet provides side impact protection.

(61) FIG. **13** shows a battery pack **1300** including a forward disposed splitter **1302** and a rearward disposed manifold **1304** for directing coolant between battery cells **1306** of the battery pack **1300** in a direction perpendicular to rows of the battery cells **1306**. The splitter **1302** has holes **1308** that vary in size from a centrally located inlet **1310**. The holes **1308** may increase in size the further the holes **1308** are away from the inlet **1310**, as shown. The manifold **1304** may have similar holes **1311** and an outlet **1312**.

(62) The cells **1306** are arranged in rows and columns, where the rows extend perpendicular to the splitter **1302** and the manifold **1304** and parallel to the direction of immersion fluid flow, which is from the splitter **1302** to the manifold **1304**. The rows include first rows that are offset from the second rows to provide a close arrangement of cells. A winding arrow is shown to illustrate direction and flow of the immersion fluid.

(63) Any of the above-described examples may be combined. Although different battery packs are shown in FIGS. **3-11B** having different features, any of the battery packs may be modified to include any of the features of FIGS. **3-13** and/or other features disclosed herein. The battery packs of FIGS. **3-11B** may include the features of FIGS. **12-13**.

(64) FIG. **14** shows a thermal runaway event detection method. The following operations may be iteratively performed. The method may begin at **1400**. At **1402**, a battery pack (or module), such as any of the battery packs disclosed herein and a swirl pot are provided and connected as described above. Although the following operations **1404**, **1406** and **1408** are shown in FIG. **14** as being performed sequentially, operations **1404**, **1406** and **1608** may be continuously performed and/or performed in parallel.

(65) At **1404**, an immersion fluid (or coolant) is circulated through the battery pack and the swirl pot. This may include circulating the immersion fluid through one or more heat exchangers, one or more chillers, one or more components, etc.

(66) At **1406**, the swirl pot performs immersion fluid deaeration. At **1408**, sensors, such as the sensors **122**, **124** are used to detect a state of the battery pack.

(67) At **1410**, one of the modules **104**, **110**, **112**, based on the sensor data, determines the state of

the battery pack and whether a thermal runaway event has occurred. If a thermal runaway event has not occurred, operation **1404** may be performed, otherwise operation **1412** may be performed.

(68) At **1412**, a countermeasure may be performed in response to detection of a thermal runaway event. This may include generating one or more alert messages via the display **220** and/or audio system **222** of FIG. 2 and/or transmitting one or more alert message to a remotely located central monitoring station or device via the transceivers **223** of FIG. 2. The countermeasures may include driving the host vehicle in a reduced feature mode to a home, a service center, a safe zone, etc. The countermeasures may include isolating, disconnecting and/or not using the cells of concern and/or shutting down one or more systems of the vehicle. The method may end at **1414**.

(69) The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

(70) Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including “connected,” “engaged,” “coupled,” “adjacent,” “next to,” “on top of,” “above,” “below,” and “disposed.” Unless explicitly described as being “direct,” when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements. As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

(71) In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

(72) In this application, including the definitions below, the term “module” or the term “controller” may be replaced with the term “circuit.” The term “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

(73) The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN), the Internet, a wide area network (WAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via

interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

(74) The term code, as used above, may include software, firmware, and/or microcode, and may refer to programs, routines, functions, classes, data structures, and/or objects. The term shared processor circuit encompasses a single processor circuit that executes some or all code from multiple modules. The term group processor circuit encompasses a processor circuit that, in combination with additional processor circuits, executes some or all code from one or more modules. References to multiple processor circuits encompass multiple processor circuits on discrete dies, multiple processor circuits on a single die, multiple cores of a single processor circuit, multiple threads of a single processor circuit, or a combination of the above. The term shared memory circuit encompasses a single memory circuit that stores some or all code from multiple modules. The term group memory circuit encompasses a memory circuit that, in combination with additional memories, stores some or all code from one or more modules.

(75) The term memory circuit is a subset of the term computer-readable medium. The term computer-readable medium, as used herein, does not encompass transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave); the term computer-readable medium may therefore be considered tangible and non-transitory. Non-limiting examples of a non-transitory, tangible computer-readable medium are nonvolatile memory circuits (such as a flash memory circuit, an erasable programmable read-only memory circuit, or a mask read-only memory circuit), volatile memory circuits (such as a static random access memory circuit or a dynamic random access memory circuit), magnetic storage media (such as an analog or digital magnetic tape or a hard disk drive), and optical storage media (such as a CD, a DVD, or a Blu-ray Disc).

(76) The apparatuses and methods described in this application may be partially or fully implemented by a special purpose computer created by configuring a general purpose computer to execute one or more particular functions embodied in computer programs. The functional blocks, flowchart components, and other elements described above serve as software specifications, which can be translated into the computer programs by the routine work of a skilled technician or programmer.

(77) The computer programs include processor-executable instructions that are stored on at least one non-transitory, tangible computer-readable medium. The computer programs may also include or rely on stored data. The computer programs may encompass a basic input/output system (BIOS) that interacts with hardware of the special purpose computer, device drivers that interact with particular devices of the special purpose computer, one or more operating systems, user applications, background services, background applications, etc.

(78) The computer programs may include: (i) descriptive text to be parsed, such as HTML (hypertext markup language), XML (extensible markup language), or JSON (JavaScript Object Notation) (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, C#, Objective-C, Swift, Haskell, Go, SQL, R, Lisp, Java®, Fortran, Perl, Pascal, Curl, OCaml, Javascript®, HTML5 (Hypertext Markup Language 5th revision), Ada, ASP (Active Server Pages), PHP (PHP: Hypertext Preprocessor), Scala, Eiffel, Smalltalk, Erlang, Ruby, Flash®, Visual Basic®, Lua, MATLAB, SIMULINK, and Python®.

Claims

1. A thermal control system comprising: a battery pack comprising a plurality of battery cells that are cylindrically-shaped and extending vertically, an outer housing configured to enclose and

immerse the plurality of battery cells in a coolant, wherein the outer housing comprises a plurality of walls including side walls, a top wall and a bottom wall, a support structure for supporting the plurality of battery cells, wherein the support structure includes a plurality of dividing members separating the plurality of battery cells, and a planar structure disposed above or below the plurality of battery cells, wherein a gap exists between a planar surface of the planar structure and a planar surface of one of the top wall and the bottom wall of the outer housing for flowing the coolant between the planar structure and the one of the top wall and the bottom wall, the planar structure and at least some of the plurality of walls direct the coolant to flow across and between the plurality of battery cells, and the planar structure and the plurality of walls direct the coolant to flow i) first across tops or bottoms of the plurality of battery cells and then laterally between the plurality of battery cells before exiting the outer housing, or ii) first laterally between the plurality of battery cells and then across the tops or the bottoms of the plurality of battery cells before exiting the outer housing; a cooling circuit comprising the battery pack and a swirl pot, wherein the swirl pot is configured to perform fluid deaeration; a sensor coupled to the swirl pot, the battery pack or another component of the cooling circuit and configured to monitoring a state of the thermal control system; and a control module is configured to detect a thermal runaway event based on an output of the sensor and, perform a countermeasure in response to detecting the thermal runaway event.

2. The thermal control system of claim 1, wherein the planar structure and the top wall direct the coolant across tops of the plurality of battery cells prior to or subsequent to flowing between the plurality of battery cells.

3. The thermal control system of claim 1, wherein the planar structure and the bottom wall direct the coolant across bottoms of the plurality of battery cells prior to or subsequent to flowing between the plurality of battery cells.

4. The thermal control system of claim 1, wherein the planar structure is configured as a splitter and receives coolant and splits the received coolant into a plurality of flow paths to direct the coolant around sides of each of the plurality of battery cells.

5. The thermal control system of claim 1, wherein the planar structure directs the coolant to flow vertically between and around each of the plurality of battery cells.

6. The thermal control system of claim 1, wherein the planar structure is configured as a manifold and receives coolant from each of the plurality of battery cells and direct the coolant to the gap between the planar structure and the one of the top wall or the bottom wall.

7. The thermal control system of claim 1, further comprising an electrical bus disposed across and electrically coupled to the plurality of battery cells, wherein: the support structure is configured to hold the electrical bus; and the support structure is configured to direct the coolant across a surface of the electrical bus.

8. The thermal control system of claim 1, wherein: the planar structure includes a plurality of holes for passage of the coolant to flow to or from each of the plurality of battery cells; the support structure includes a plurality of feet, wherein open areas between the plurality of feet are respectively over the plurality of holes for passage of the coolant to flow to or from each of the plurality of battery cells; and gaps between the plurality of feet allow the coolant to flow between the plurality of feet and along sides of the plurality of battery cells.

9. A thermal control system comprising: a battery pack comprising a plurality of battery cells, an outer housing configured to enclose and immerse the plurality of battery cells in a coolant, wherein the outer housing comprises a plurality of walls including side walls, a top wall and a bottom wall, a support structure for supporting the plurality of battery cells, wherein the support structure includes a plurality of dividing members separating the plurality of battery cells, and a planar structure disposed above or below the plurality of battery cells, wherein the planar structure directs the coolant to flow vertically between and around each of the plurality of battery cells, a gap exists between a planar surface of the planar structure and a planar surface of one of the top wall and the

bottom wall of the outer housing for flowing the coolant between the planar structure and the one of the top wall and the bottom wall, the planar structure and at least some of the plurality of walls direct the coolant to flow across and between the plurality of battery cells, the planar structure includes a first plurality of holes for passage of the coolant to flow to or from each of the plurality of battery cells, the support structure includes a second plurality of holes, the first plurality of holes are larger than the second plurality of holes, and protruding members of the support structure are received respectively within the first plurality of holes of the planar structure such that the protruding members reduce sizes of pass through openings in the planar structure from being sizes of the first plurality of holes to being sizes of the second plurality of holes, and as a result restricting the flow of coolant through the planar structure; a cooling circuit comprising the battery pack and a swirl pot, wherein the swirl pot is configured to perform fluid deaeration; a sensor coupled to the swirl pot, the battery pack or another component of the cooling circuit and configured to monitoring a state of the thermal control system; and a control module is configured to detect a thermal runaway event based on an output of the sensor and, perform a countermeasure in response to detecting the thermal runaway event.

10. The thermal control system of claim 1, wherein: the support structure includes a plurality of orifices that are arranged for flow of the coolant to or from the plurality of battery cells; and each portion of the support structure surrounding a respective one of the plurality of orifices is formed of a material that melts during a thermal runaway event of a corresponding one of the plurality of battery cells.

11. The thermal control system of claim 1, further comprising the support structure includes a plurality of protruding elements extending in gaps between the plurality of battery cells.

12. The thermal control system of claim 11, wherein the plurality of protruding elements comprise at least one of spiral-shaped elements, disks, rods, or trees, which provide fluid flow resistance for the coolant flowing through the gaps.

13. A cooling system comprising: the thermal control system of claim 1; and a pump configured to pump coolant to and from the battery pack, wherein the control module is configured to control the pump to circulate the coolant through the battery pack.

14. A cooling system comprising: the thermal control system of claim 1; at least one of a cell monitoring module, the control module, a battery disconnect module, or a motor; and a plurality of conduits connected to and directing the coolant between the battery pack and the at least one of the cell monitoring module, the control module, the battery disconnect module, or the motor.

15. A cooling system comprising: the thermal control system of claim 1; and at least one of a cell monitoring module, the control module or a battery disconnect module disposed on a wall of the battery pack and contacting the coolant in the battery pack.

16. A thermal control system comprising: a battery pack comprising a plurality of battery cells, an outer housing configured to enclose and immerse the plurality of battery cells in a coolant, wherein the outer housing comprises a plurality of walls including side walls, a top wall and a bottom wall, a support structure for supporting the plurality of battery cells, wherein the support structure includes a plurality of dividing members separating the plurality of battery cells, and a planar structure disposed above or below the plurality of battery cells, wherein a gap exists between a planar surface of the planar structure and a planar surface of one of the top wall and the bottom wall of the outer housing for flowing the coolant between the planar structure and the one of the top wall and the bottom wall, and the planar structure and at least some of the plurality of walls direct the coolant to flow across and between the plurality of battery cells; a cooling circuit comprising the battery pack and a swirl pot, wherein the swirl pot is configured to perform fluid deaeration; a sensor coupled to the swirl pot, the battery pack or another component of the cooling circuit and configured to monitoring a state of the thermal control system; and a control module is configured to detect a thermal runaway event based on an output of the sensor and, perform a countermeasure in response to detecting the thermal runaway event.

17. The thermal control system of claim 1, wherein: the planar structure is immersed in the coolant and cools the tops or the bottoms of the plurality of battery cells; the planar structure extends at least one of i) horizontally and ii) perpendicular to a longitudinal end-to-end extending direction of each of the plurality of battery cells; and the plurality of battery cells are disposed on a first side of the planar structure and the coolant flows across a second side of the planar structure opposite the first side.

18. The thermal control system of claim 1, wherein: the coolant flows i) first across all of the tops or all of the bottoms of the plurality of battery cells and then laterally between the plurality of battery cells before exiting the outer housing, or ii) first laterally between the plurality of battery cells and then across all of the tops or all of the bottoms of the plurality of battery cells before exiting the outer housing; and the direction of coolant flow between the plurality of battery cells is at least one of i) horizontal and ii) perpendicular to a longitudinal end-to-end extending direction of each of the plurality of battery cells.
